

8-Bit Function

//Verilog code

`timescale 1ns / 1ps

module function_prog (

input wire [3:0]a, b, c, d, e, f, g, h,

output wire [7:0] y);

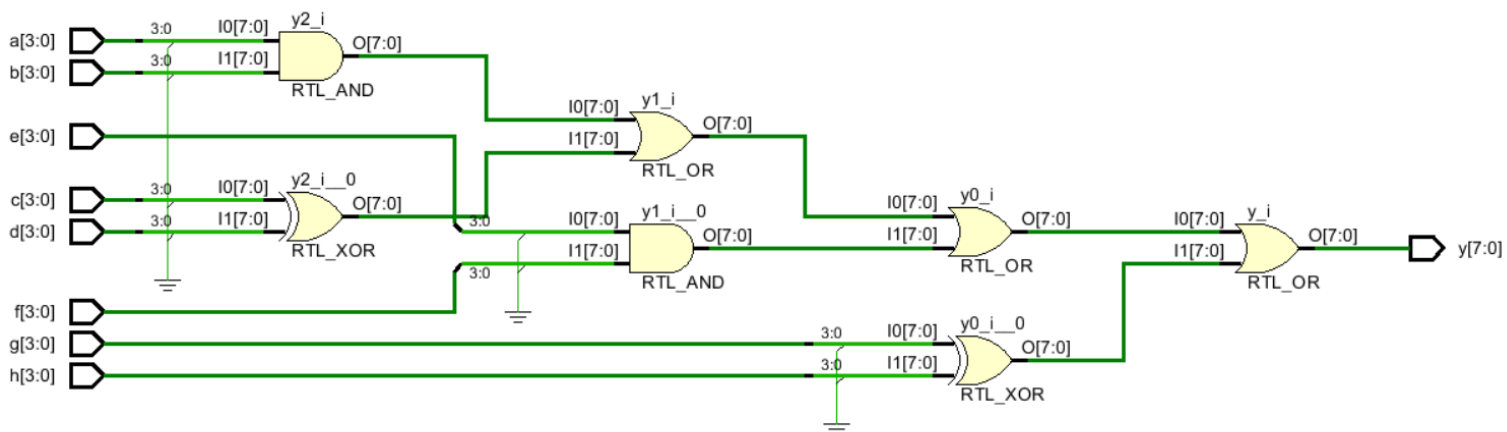
assign y = a & b | c ^ d | e & f | g ^ h;

endmodule

1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	8	0	53200	0.02
LUT as Logic	8	0	53200	0.02
LUT as Memory	0	0	17400	0.00
Slice Registers	0	0	106400	0.00
Register as Flip Flop	0	0	106400	0.00
Register as Latch	0	0	106400	0.00
F7 Muxes	0	0	26600	0.00
F8 Muxes	0	0	13300	0.00

RTL Analysis schematic:



Look Up Tables:

Cell Properties

■ y_OBUF[0]_inst_i_2

I3	I2	I1	I0	O=I0 & I1 + I2 & !I3 + !I2 & I3
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Table 1: LUT for all LUT4

Cell Properties

■ y_OBUF[0]_inst_i_1

I4	I3	I2	I1	I0	O=I4 + I2 & I3 + I0 & !I1 & !I4 + !I0 & I1 & !I4
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	1
0	0	0	1	1	0
0	0	1	0	0	0
0	0	1	0	1	1
0	0	1	1	0	1
0	0	1	1	1	0
0	1	0	0	0	0
0	1	0	0	1	1
0	1	0	1	0	1
0	1	0	1	1	0
0	1	1	0	0	1
0	1	1	0	1	1
0	1	1	1	0	1
0	1	1	1	1	1
1	0	0	0	0	1
1	0	0	0	1	1
1	0	0	1	0	1
1	0	0	1	1	1
1	0	1	0	0	1
1	0	1	0	1	1
1	0	1	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	0	0	1	1
1	1	0	1	0	1
1	1	0	1	1	1
1	1	1	0	0	1
1	1	1	0	1	1
1	1	1	1	0	1
1	1	1	1	1	1

Table 2: LUT for all LUT5